

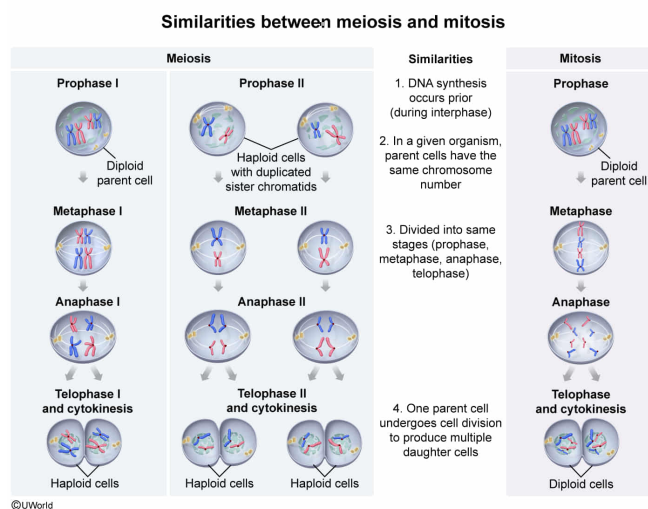
In humans, meiosis and mitosis are most similar with respect to:

- ☐ A. the genetic makeup of daughter cells. [7%]
- ☐ B. the synapsis of homologous chromosomes. [17%]
- ☐ C. the total number of cell divisions. [3%]
- ✓ ☒ D. the parent cell chromosome number. [71%]

Correct

71% answered correctly

Explanation:



Meiosis is a reproductive process that results in the formation of **gametes** (ie, egg, sperm). **Mitosis** is also a reproductive process that occurs in somatic (non-sex) cells for organism **growth and repair**.

There are some similarities between meiosis and mitosis:

- DNA synthesis occurs prior to cell division (ie, during the **S phase** of **interphase**).
- In a given organism, **parent cells have the same chromosome number** (eg, diploid or $2n$ in humans).
- Both are divided into the same general stages (ie, prophase, metaphase, anaphase, telophase).
- One parent cell divides to produce multiple daughter cells.

However, there are also differences between meiosis and mitosis:

- Four daughter cells result from meiosis whereas two daughter cells result from mitosis.

- **Haploid cells result** from meiosis whereas **diploid cells result** from mitosis.
- Meiosis occurs to form gametes whereas mitosis occurs for growth and repair.
- Daughter cells differ from the original parent cell in meiosis whereas daughter cells are genetically identical to the original parent cell in mitosis **(Choice A)**.
- **Synapsis** (close association) of **homologous chromosomes** occurs during meiosis but not during mitosis **(Choice B)**.
- Two rounds of cell division occur during meiosis whereas only one round of cell division occurs during mitosis **(Choice C)**.

Accordingly, meiosis and mitosis in a given organism are similar with respect to the **parent cell chromosome number**. Human diploid cells contain 46 chromosomes, so the diploid parent cells that undergo meiosis and the diploid parent cells that undergo mitosis **both contain 46 chromosomes**.

Educational objective:

Differences between meiosis and mitosis include resulting cell number and ploidy, function, similarity to parent cell, synapsis, and number of cell divisions. Similarities between these two processes include preceding DNA synthesis, parent cell chromosome number, general stages, and parent cell number.

Time spent:0 sec

Subject:
Biology

QID:403078

System:
Genetics and Evolution